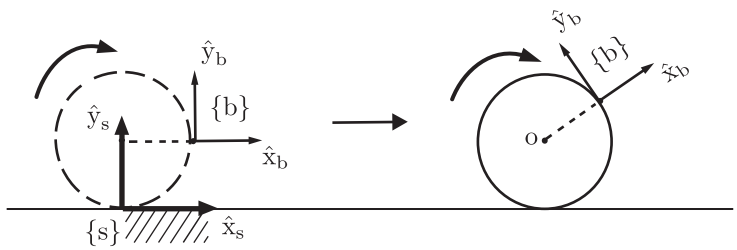


Figure 5.14: A rolling wheel.

screws for the joint axes. Compact closed-form expressions for the columns of the Jacobian are also obtained because taking derivatives of matrix exponentials is particularly straightforward.

There is extensive literature on the singularity analysis of 6R open chains. In addition to the three cases presented in this chapter, other cases are examined in [122] and in exercises at the end of this chapter, including the case when some of the revolute joints are replaced by prismatic joints. Many of the mathematical techniques and analyses used in open chain singularity analysis can also be used to determine the singularities of parallel mechanisms, which are the subject of Chapter 7.

The concept of a robot's manipulability was first formulated in a quantitative way by Yoshikawa [195]. There is now a vast literature on the manipulability analysis of open chains, see, e.g., [75, 134].

5.8 Exercises

Exercise 5.1 A wheel of unit radius is rolling to the right at a rate of 1 rad/s (see Figure 5.14; the dashed circle shows the wheel at $t = 0$).

(a) Find the spatial twist $\mathcal{V}_s(t)$ as a function of t.

(b) Find the linear velocity of the {b}-frame origin expressed in {s}-frame coordinates.

Exercise 5.2 The 3R planar open chain of Figure 5.15(a) is shown in its zero position.

(a) Suppose that the last link must apply a wrench corresponding to a force of 5 N in the $\hat{x}_s$-direction of the {s} frame, with zero component in the $\hat{y}_s$-direction and zero moment about the $\hat{z}_s$ axis. What torques should be

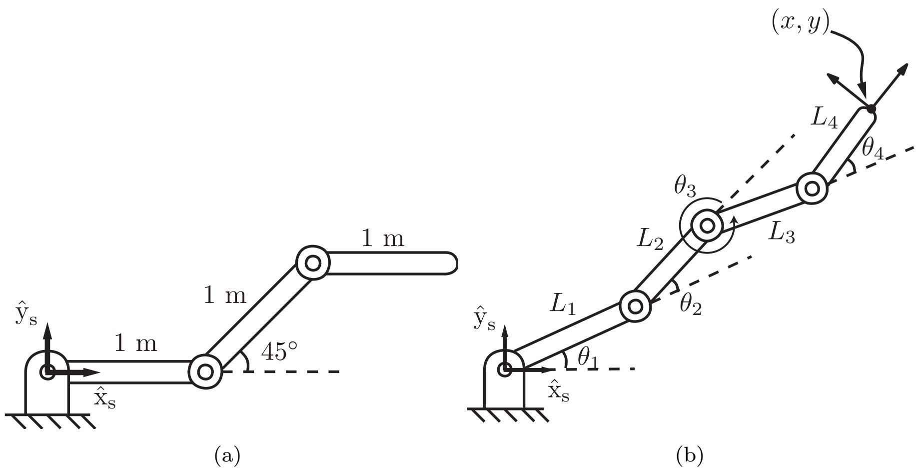


Figure 5.15: (a) A 3R planar open chain. The length of each link is 1 m. (b) A 4R planar open chain.

applied at each joint?

(b) Suppose that the last link must apply a force of 5 N in the $\hat{y}_s$-direction, with zero components in the other wrench directions. What torques should be applied at each joint?

Exercise 5.3 Answer the following questions for the 4R planar open chain of Figure 5.15(b).

(a) For the forward kinematics of the form

$$T(\theta) = e^{[\mathcal{S}_1]\theta_1} e^{[\mathcal{S}_2]\theta_2} e^{[\mathcal{S}_3]\theta_3} e^{[\mathcal{S}_4]\theta_4} M,$$

write down $M \in SE(2)$ and each $\mathcal{S}_i = (\omega_{zi}, v_{xi}, v_{yi}) \in \mathbb{R}^3$.

(b) Write down the body Jacobian.

(c) Suppose that the chain is in static equilibrium at the configuration $\theta_1 = \theta_2 = 0, \theta_3 = \pi/2, \theta_4 = -\pi/2$ and that a force $f = (10, 10, 0)$ and a moment $m = (0, 0, 10)$ are applied to the tip (both f and m are expressed with respect to the fixed frame). What are the torques experienced at each joint?

(d) Under the same conditions as (c), suppose that a force $f = (-10, 10, 0)$ and a moment $m = (0, 0, -10)$, also expressed in the fixed frame, are applied to the tip. What are the torques experienced at each joint?

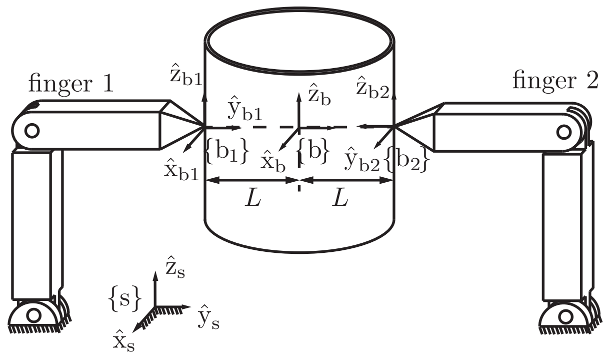


Figure 5.16: Two fingers grasping a can.

(e) Find all kinematic singularities for this chain.

Exercise 5.4 Figure 5.16 shows two fingers grasping a can. Frame {b} is attached to the center of the can. Frames {b$_1$} and {b$_2$} are attached to the can at the two contact points as shown. The force $f_1 = (f_{1,x}, f_{1,y}, f_{1,z})$ is the force applied by fingertip 1 to the can, expressed in {b$_1$} coordinates. Similarly, $f_2 = (f_{2,x}, f_{2,y}, f_{2,z})$ is the force applied by fingertip 2 to the can, expressed in {b$_2$} coordinates.

(a) Assume that the system is in static equilibrium, and find the total wrench $\mathcal{F}_b$ applied by the two fingers to the can. Express your result in {b} coordinates.

(b) Suppose that $\mathcal{F}_{\text{ext}}$ is an arbitrary external wrench applied to the can ($\mathcal{F}_{\text{ext}}$ is also expressed in frame-{b} coordinates). Find all $\mathcal{F}_{\text{ext}}$ that cannot be resisted by the fingertip forces.

Exercise 5.5 Referring to Figure 5.17, a rigid body, shown at the top right, rotates about the point (L, L) with angular velocity $\dot{\theta} = 1$.

(a) Find the position of point P on the moving body relative to the fixed reference frame {s} in terms of θ.

(b) Find the velocity of point P in terms of the fixed frame.

(c) What is T_{sb}, the configuration of frame {b}, as seen from the fixed frame {s}?

(d) Find the twist of T_{sb} in body coordinates.

(e) Find the twist of T_{sb} in space coordinates.

(f) What is the relationship between the twists from (d) and (e)?

(g) What is the relationship between the twist from (d) and $\dot{P}$ from (b)?

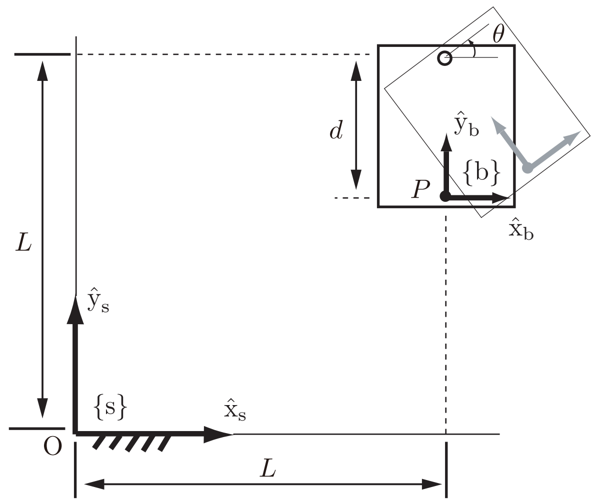


Figure 5.17: A rigid body rotating in the plane.

(h) What is the relationship between the twist from (e) and $\dot{P}$ from (b)?

Exercise 5.6 Figure 5.18 shows a design for a new amusement park ride. A rider sits at the location indicated by the moving frame {b}. The fixed frame {s} is attached to the top shaft as shown. The dimensions indicated in the figure are $R = 10$ m and $L = 20$ m, and the two joints each rotate at a constant angular velocity of 1 rad/s.

(a) Suppose $t = 0$ at the instant shown in the figure. Find the linear velocity v_b and angular velocity ω_b of the rider as functions of time t. Express your answer in frame-{b} coordinates.

(b) Let p be the linear coordinates expressing the position of the rider in {s}. Find the linear velocity $\dot{p}(t)$.

Exercise 5.7 The RRP robot in Figure 5.19 is shown in its zero position.

(a) Write down the screw axes in the space frame. Evaluate the forward kinematics when $\theta = (90^\circ, 90^\circ, 1)$. Hand-draw or use a computer to show the arm and the end-effector frame in this configuration. Obtain the space Jacobian J_s for this configuration.

(b) Write down the screw axes in the end-effector body frame. Evaluate the forward kinematics when $\theta = (90^\circ, 90^\circ, 1)$ and confirm that you get the same result as in part (a). Obtain the body Jacobian J_b for this configu-

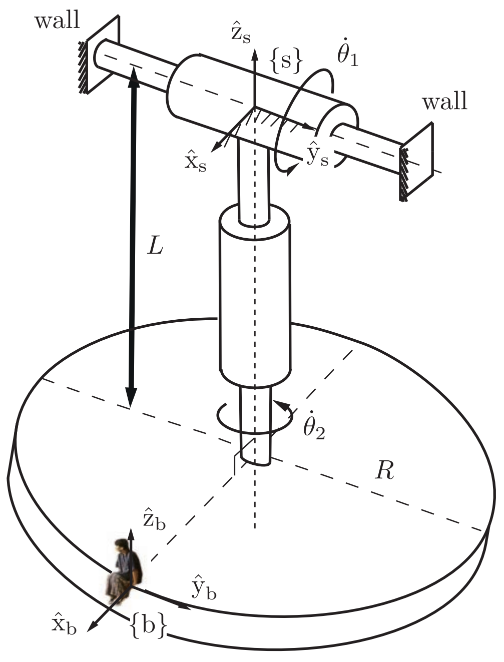


Figure 5.18: A new amusement park ride.

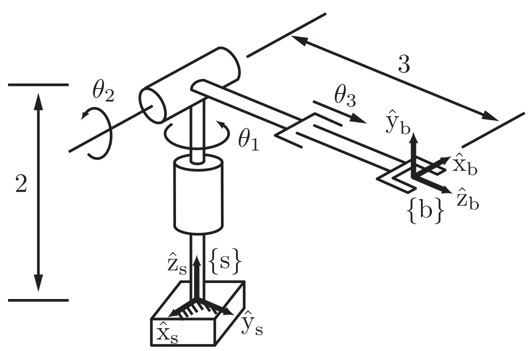


Figure 5.19: RRP robot shown in its zero position.

ration.

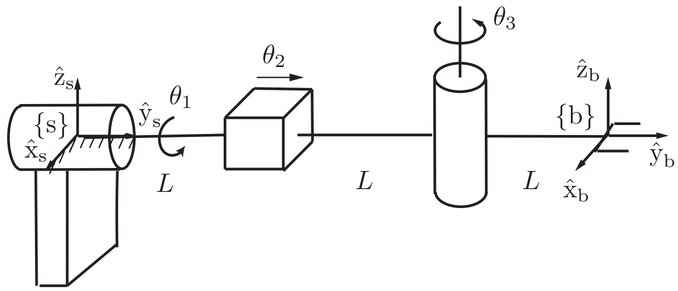


Figure 5.20: RPR robot.

Exercise 5.8 The RPR robot of Figure 5.20 is shown in its zero position. The fixed and end-effector frames are respectively denoted {s} and {b}.

(a) Find the space Jacobian $J_s(\theta)$ for arbitrary configurations $\theta \in \mathbb{R}^3$.

(b) Assume the manipulator is in its zero position. Suppose that an external force $f \in \mathbb{R}^3$ is applied to the {b} frame origin. Find all the directions in which f can be resisted by the manipulator with $\tau = 0$.

Exercise 5.9 Find the kinematic singularities of the 3R wrist given the forward kinematics

$$R = e^{[\hat{\omega}_1]\theta_1} e^{[\hat{\omega}_2]\theta_2} e^{[\hat{\omega}_3]\theta_3},$$

where $\hat{\omega}_1 = (0, 0, 1)$, $\hat{\omega}_2 = (1/\sqrt{2}, 0, 1/\sqrt{2})$, and $\hat{\omega}_3 = (1, 0, 0)$.

Exercise 5.10 In this exercise, for an n-link open chain we derive the analytic Jacobian corresponding to the exponential coordinates on $SO(3)$.

(a) Given an $n \times n$ matrix $A(t)$ parametrized by t that is also differentiable with respect to t, its exponential $X(t) = e^{A(t)}$ is then an $n \times n$ matrix that is always nonsingular. Prove the following:

$$\begin{aligned} X^{-1}\dot{X} &= \int_0^1 e^{-A(t)s} \dot{A}(t) e^{A(t)s} ds, \\ \dot{X}X^{-1} &= \int_0^1 e^{A(t)s} \dot{A}(t) e^{-A(t)s} ds. \end{aligned}$$

(Hint: The formula

$$\frac{d}{d\epsilon}e^{(A+\epsilon B)t}|_{\epsilon=0} = \int_0^t e^{As}Be^{A(t-s)}ds$$

may be useful.)

(b) Use the result above to show that, for $r(t) \in \mathbb{R}^3$ and $R(t) = e^{[r(t)]}$, the angular velocity in the body frame, $[\omega_b] = R^{\mathrm{T}}\dot{R}$, is related to $\dot{r}$ by

$$\begin{aligned} \omega_b &= A(r)\dot{r}, \\ A(r) &= I - \frac{1-\cos\|r\|}{\|r\|^2}[r] + \frac{\|r\| - \sin\|r\|}{\|r\|^3}[r]^2. \end{aligned}$$

(c) Derive the corresponding formula relating the angular velocity in the space frame, $[\omega_s] = \dot{R}R^{\mathrm{T}}$, to $\dot{r}$.

Exercise 5.11 The spatial 3R open chain of Figure 5.21 is shown in its zero position. Let p be the coordinates of the origin of $\{b\}$ expressed in $\{s\}$.

(a) In its zero position, suppose we wish to make the end-effector move with linear velocity $\dot{p} = (10, 0, 0)$. Is this motion possible? If so, what are the required input joint velocities $\dot{\theta}_1, \dot{\theta}_2$, and $\dot{\theta}_3$?

(b) Suppose that the robot is in the configuration $\theta_1 = 0, \theta_2 = 45°, \theta_3 = -45°$. Assuming static equilibrium, suppose that we wish to generate an end-effector force $f_b = (10, 0, 0)$, where f_b is expressed with respect to the end-effector frame $\{b\}$. What are the required input joint torques τ_1, τ_2, and τ_3?

(c) Under the same conditions as in (b), suppose that we now seek to generate an end-effector moment $m_b = (10, 0, 0)$, where m_b is expressed with respect to the end-effector frame $\{b\}$. What are the required input joint torques τ_1, τ_2, τ_3?

(d) Suppose that the maximum allowable torques for each joint motor are

$$\|\tau_1\| \le 10, \quad \|\tau_2\| \le 20, \qquad \text{and} \qquad \|\tau_3\| \le 5.$$

In the zero position, what is the maximum force that can be applied by the tip in the end-effector-frame $\hat{\mathrm{x}}$-direction?

Exercise 5.12 The RRRP chain of Figure 5.22 is shown in its zero position. Let p be the coordinates of the origin of $\{b\}$ expressed in $\{s\}$.

(a) Determine the body Jacobian $J_b(\theta)$ when $\theta_1 = \theta_2 = 0, \theta_3 = \pi/2, \theta_4 = L$.

(b) Find $\dot{p}$ when $\theta_1 = \theta_2 = 0, \theta_3 = \pi/2, \theta_4 = L$ and $\dot{\theta}_1 = \dot{\theta}_2 = \dot{\theta}_3 = \dot{\theta}_4 = 1$.

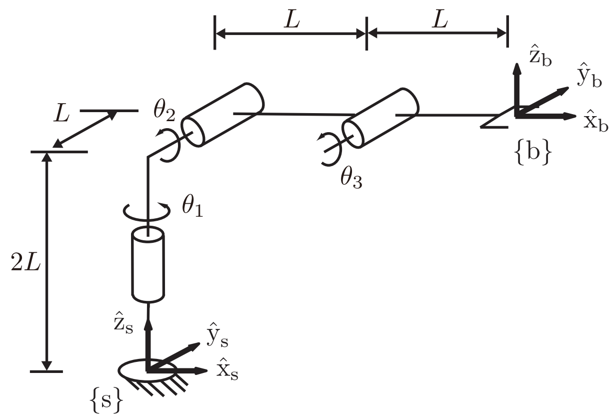


Figure 5.21: A spatial 3R open chain.

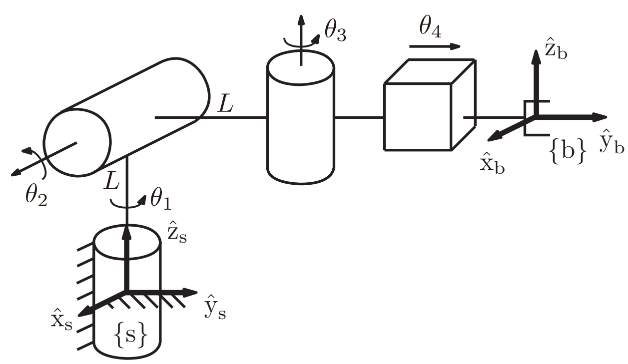


Figure 5.22: An RRRP spatial open chain.

Exercise 5.13 For the 6R spatial open chain of Figure 5.23,

(a) Determine its space Jacobian $J_s(\theta)$.

(b) Find its kinematic singularities. Explain each singularity in terms of the alignment of the joint screws and of the directions in which the end-effector loses one or more degrees of freedom of motion.

Exercise 5.14 Show that a six-dof spatial open chain is at a kinematic sin-

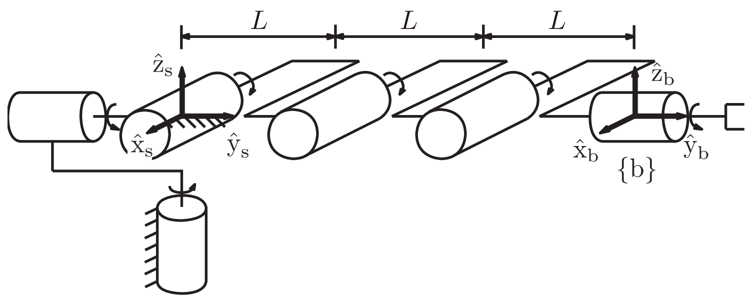


Figure 5.23: Singularities of a 6R open chain.

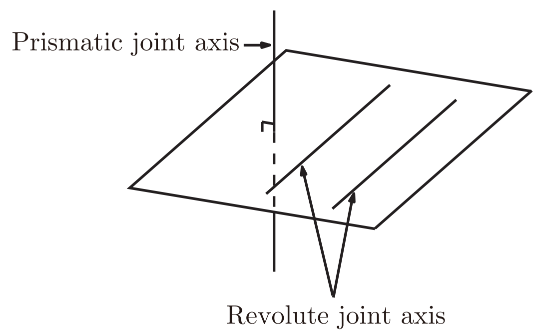


Figure 5.24: A kinematic singularity involving prismatic and revolute joints.

gularity when any two of its revolute joint axes are parallel, and any prismatic joint axis is normal to the plane spanned by the two parallel revolute joint axes (see Figure 5.24).

Exercise 5.15 The spatial PRRRRP open chain of Figure 5.25 is shown in its zero position.

(a) At the zero position, find the first three columns of the space Jacobian.
(b) Find all configurations for which the first three columns of the space Jacobian become linearly dependent.
(c) Suppose that the chain is in the configuration $\theta_1 = \theta_2 = \theta_3 = \theta_5 = \theta_6 = 0$, $\theta_4 = 90°$. Assuming static equilibrium, suppose that a pure force $f_b = (10, 0, 10)$, where f_b is expressed in terms of the end-effector frame, is applied to the origin of the end-effector frame. Find the torques τ_1, τ_2,

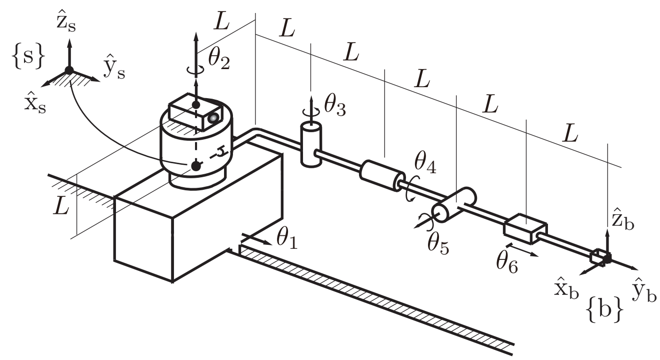


Figure 5.25: A spatial PRRRRP open chain.

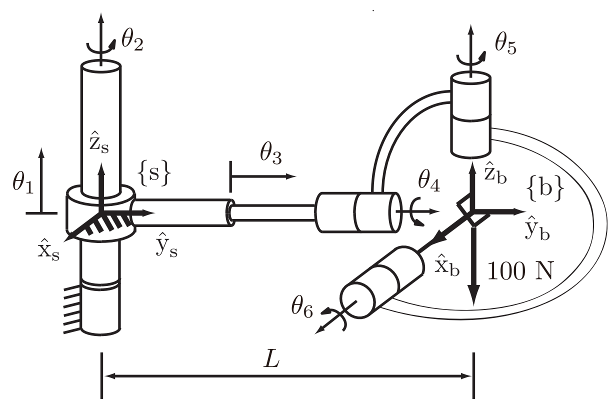


Figure 5.26: A PRPRRR spatial open chain.

and τ_3 experienced at the first three joints.

Exercise 5.16 Consider the PRPRRR spatial open chain of Figure 5.26, shown in its zero position. The distance from the origin of the fixed frame to the origin of the end-effector frame at the home position is L.

(a) Determine the first three columns of the space Jacobian J_s.
(b) Determine the last two columns of the body Jacobian J_b.
(c) For what value of L is the home position a singularity?
(d) In the zero position, what joint forces and torques τ must be applied in order to generate a pure end-effector force of 100 N in the $-\hat{z}_b$-direction?

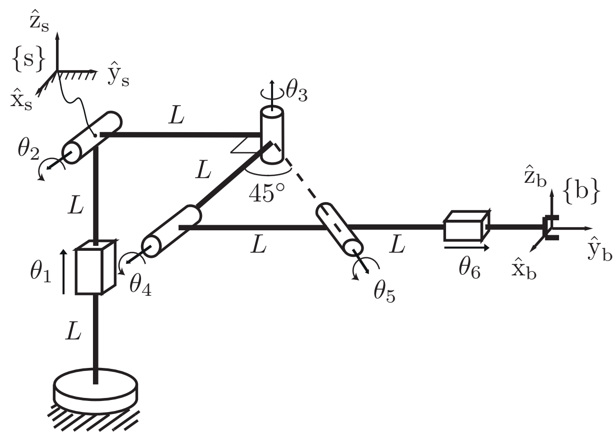


Figure 5.27: A PRRRRP robot.

Exercise 5.17 The PRRRRP robot of Figure 5.27 is shown in its zero position.
(a) Find the first three columns of the space Jacobian $J_s(\theta)$.
(b) Assuming the robot is in its zero position and $\dot{\theta} = (1, 0, 1, -1, 2, 0)$, find the spatial twist $\mathcal{V}_s$.
(c) Is the zero position a kinematic singularity? Explain your answer.

Exercise 5.18 The six-dof RRPRPR open chain of Figure 5.28 has a fixed frame {s} and an end-effector frame {b} attached as shown. At its zero position, joint axes 1, 2, and 6 lie in the ŷ–ẑ-plane of the fixed frame, and joint axis 4 is aligned along the fixed-frame x̂-axis.
(a) Find the first three columns of the space Jacobian $J_s(\theta)$.
(b) At the zero position, let $\dot{\theta} = (1, 0, 1, -1, 2, 0)$. Find the spatial twist $\mathcal{V}_s$.
(c) Is the zero position a kinematic singularity? Explain your answer.

Exercise 5.19 The spatial PRRRRP open chain of Figure 5.29 is shown in its zero position.
(a) Determine the first four columns of the space Jacobian $J_s(\theta)$.
(b) Determine whether the zero position is a kinematic singularity.
(c) Calculate the joint forces and torques required for the tip to apply the following end-effector wrenches:
   (i) $\mathcal{F}_s = (0, 1, -1, 1, 0, 0)$.

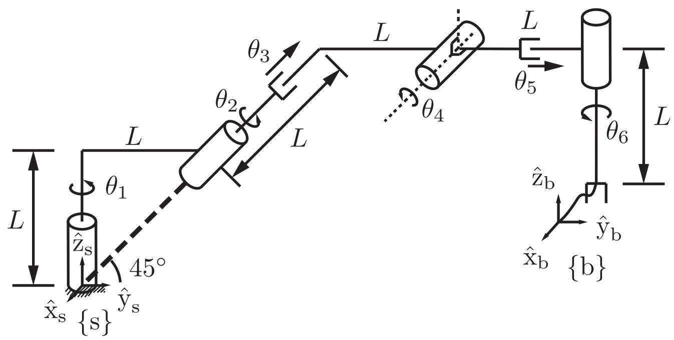


Figure 5.28: An RRPRPR open chain shown at its zero position.

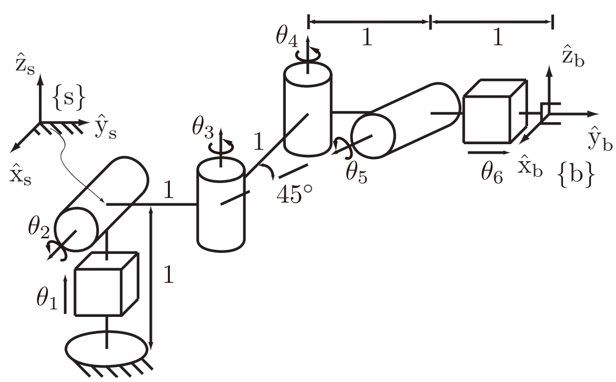


Figure 5.29: A spatial PRRRRP open chain with a skewed joint axis.

(ii) $\mathcal{F}_s = (1, -1, 0, 1, 0, -1)$.

Exercise 5.20 The spatial RRPRRR open chain of Figure 5.30 is shown in its zero position.

(a) For the fixed frame {0} and tool (end-effector) frame {t} as shown, express the forward kinematics in the product of exponentials form

$$T(\theta) = e^{[\mathcal{S}_1]\theta_1} e^{[\mathcal{S}_2]\theta_2} e^{[\mathcal{S}_3]\theta_3} e^{[\mathcal{S}_4]\theta_4} e^{[\mathcal{S}_5]\theta_5} e^{[\mathcal{S}_6]\theta_6} M.$$

(b) Find the first three columns of the space Jacobian $J_s(\theta)$.

(c) Suppose that the fixed frame {0} is moved to another location {0′} as shown in the figure. Find the first three columns of the space Jacobian

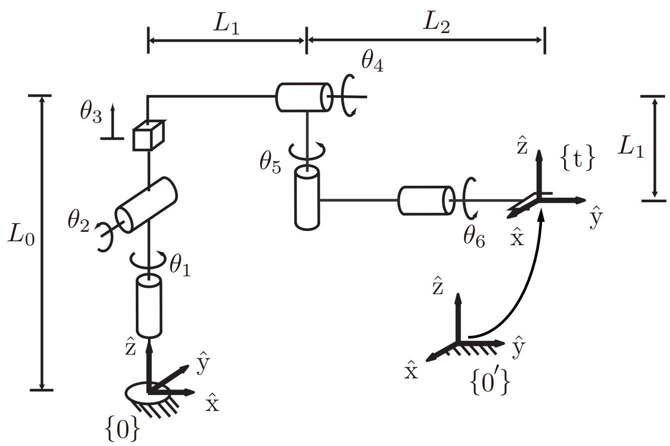


Figure 5.30: A spatial RRPRRR open chain.

$J_s(\theta)$ with respect to this new fixed frame.

(d) Determine whether the zero position is a kinematic singularity and, if so, provide a geometric description in terms of the joint screw axes.

Exercise 5.21 Figure 5.31 shows an RRPRRR exercise robot used for stroke patient rehabilitation.

(a) Assume the manipulator is in its zero position. Suppose that $M_{0c} \in SE(3)$ is the displacement from frame $\{0\}$ to frame $\{c\}$ and $M_{ct} \in SE(3)$ is the displacement from frame $\{c\}$ to frame $\{t\}$. Express the forward kinematics T_{0t} in the form

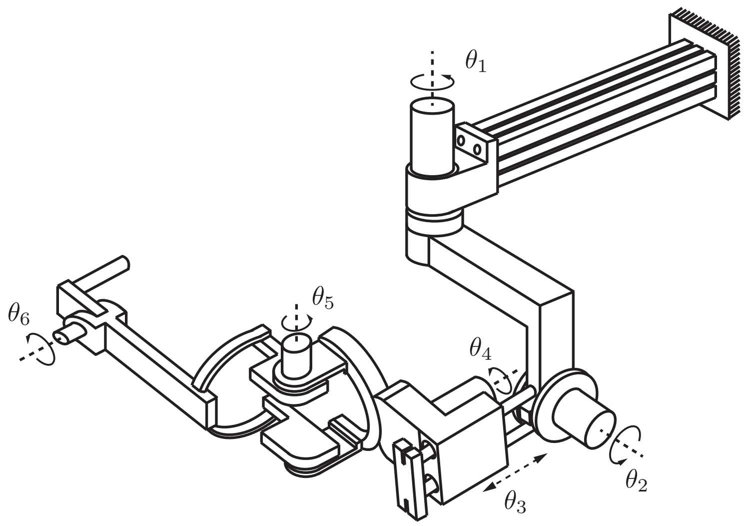
$$T_{0t} = e^{[\mathcal{A}_1]\theta_1} e^{[\mathcal{A}_2]\theta_2} M_{0c} e^{[\mathcal{A}_3]\theta_3} e^{[\mathcal{A}_4]\theta_4} M_{ct} e^{[\mathcal{A}_5]\theta_5} e^{[\mathcal{A}_6]\theta_6}.$$

Find $\mathcal{A}_2, \mathcal{A}_4$, and $\mathcal{A}_5$.

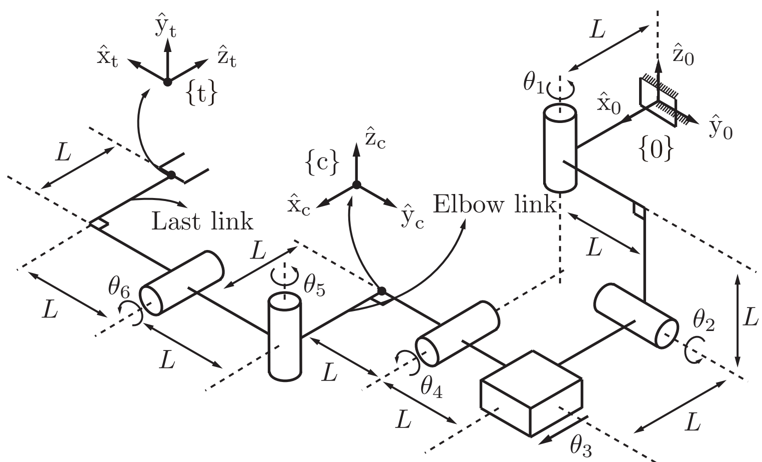
(b) Suppose that $\theta_2 = 90^\circ$ and all the other joint variables are fixed at zero. Set the joint velocities to $(\dot{\theta}_1, \dot{\theta}_2, \dot{\theta}_3, \dot{\theta}_4, \dot{\theta}_5, \dot{\theta}_6) = (1, 0, 1, 0, 0, 1)$, and find the spatial twist $\mathcal{V}_s$ in frame-$\{0\}$ coordinates.

(c) Is the configuration described in part (b) a kinematic singularity? Explain your answer.

(d) Suppose that a person now operates the rehabilitation robot. At the configuration described in part (b), a wrench $\mathcal{F}_{\text{elbow}}$ is applied to the elbow link, and a wrench $\mathcal{F}_{\text{tip}}$ is applied to the last link. Both $\mathcal{F}_{\text{elbow}}$ and $\mathcal{F}_{\text{tip}}$ are expressed in frame $\{0\}$ coordinates and are given by $\mathcal{F}_{\text{elbow}} =$



(a) Rehabilitation robot ARMin III [123]. Figure courtesy of ETH Zürich.



(b) Kinematic model of the ARMin III.

Figure 5.31: The ARMin III rehabilitation robot.

$(1, 0, 0, 0, 0, 1)$ and $\mathcal{F}_{\text{tip}} = (0, 1, 0, 1, 1, 0)$. Find the joint forces and torques

τ that must be applied for the robot to maintain static equilibrium.

Exercise 5.22 Consider an n-link open chain, with reference frames attached to each link. Let

$$T_{0k} = e^{[\mathcal{S}_1]\theta_1} \cdots e^{[\mathcal{S}_k]\theta_k} M_k, \qquad k = 1, \ldots, n$$

be the forward kinematics up to link frame $\{k\}$. Let $J_s(\theta)$ be the space Jacobian for T_{0n}; $J_s(\theta)$ has columns J_{si} as shown below:

$$J_s(\theta) = \left[\begin{array}{ccc} J_{s1}(\theta) & \cdots & J_{sn}(\theta) \end{array} \right].$$

Let $[\mathcal{V}_k] = \dot{T}_{0k} T_{0k}^{-1}$ be the twist of link frame $\{k\}$ in fixed frame $\{0\}$ coordinates.

(a) Derive explicit expressions for $\mathcal{V}_2$ and $\mathcal{V}_3$.

(b) On the basis of your results from (a), derive a recursive formula for $\mathcal{V}_{k+1}$ in terms of $\mathcal{V}_k$, $J_{s1}, \ldots, J_{s,k+1}$, and $\dot{\theta}$.

Exercise 5.23 Write a program that allows the user to enter the link lengths L_1 and L_2 of a 2R planar robot (Figure 5.32) and a list of robot configurations (each defined by the joint angles (θ_1, θ_2)) and plots the manipulability ellipse at each of those configurations. The program should plot the arm (as two line segments) at each configuration and the manipulability ellipse centered at the endpoint of the arm. Choose the same scaling for all the ellipses so that they can be easily visualized (e.g., the ellipse should usually be shorter than the arm but not so small that you cannot easily see it). The program should also print the three manipulability measures μ_1, μ_2, and μ_3 for each configuration.

(a) Choose $L_1 = L_2 = 1$ and plot the arm and its manipulability ellipse at the four configurations $(-10^\circ, 20^\circ), (60^\circ, 60^\circ), (135^\circ, 90^\circ)$, and $(190^\circ, 160^\circ)$. At which of these configurations does the manipulability ellipse appear most isotropic? Does this agree with the manipulability measures calculated by the program?

(b) Does the ratio of the length of the major axis of the manipulability ellipse and the length of the minor axis depend on θ_1? On θ_2? Explain your answers.

(c) Choose $L_1 = L_2 = 1$. Hand-draw the following: the arm at $(-45^\circ, 90^\circ)$; the endpoint linear velocity vector arising from $\dot{\theta}_1 = 1$ rad/s and $\dot{\theta}_2 = 0$; the endpoint linear velocity vector arising from $\dot{\theta}_1 = 0$ and $\dot{\theta}_2 = 1$ rad/s; and the vector sum of these two vectors to get the endpoint linear velocity when $\dot{\theta}_1 = 1$ rad/s and $\dot{\theta}_2 = 1$ rad/s.

Exercise 5.24 Modify the program in the previous exercise to plot the force

Figure 5.32: Left: The 2R robot arm. Right: The arm at four different configurations.

ellipse. Demonstrate it at the same four configurations as in the first part of the previous exercise.

Exercise 5.25 The kinematics of the 6R UR5 robot are given in Section 4.1.2.

(a) Give the numerical space Jacobian J_s when all joint angles are $\pi/2$. Separate the Jacobian matrix into an angular velocity portion J_ω (the joint rates act on the angular velocity) and a linear velocity portion J_v (the joint rates act on the linear velocity).

(b) For this configuration, calculate the directions and lengths of the principal semi-axes of the three-dimensional angular-velocity manipulability ellipsoid (based on J_ω) and the directions and lengths of the principal semi-axes of the three-dimensional linear-velocity manipulability ellipsoid (based on J_v). Comment on why it is usually preferred to use the body Jacobian instead of the space Jacobian for the manipulability ellipsoid.

(c) For this configuration, calculate the directions and lengths of the principal semi-axes of the three-dimensional moment (torque) force ellipsoid (based on J_ω) and the directions and lengths of the principal semi-axes of the three-dimensional linear force ellipsoid (based on J_v).

Exercise 5.26 The kinematics of the 7R WAM robot are given in Section 4.1.3.

(a) Give the numerical body Jacobian J_b when all joint angles are $\pi/2$. Separate the Jacobian matrix into an angular-velocity portion J_ω (the joint rates act on the angular velocity) and a linear-velocity portion J_v (the joint rates act on the linear velocity).

(b) For this configuration, calculate the directions and lengths of the principal semi-axes of the three-dimensional angular-velocity manipulability ellipsoid (based on J_ω) and the directions and lengths of the principal semi-axes of the three-dimensional linear-velocity manipulability ellipsoid (based on J_v).

(c) For this configuration, calculate the directions and lengths of the principal semi-axes of the three-dimensional moment (torque) force ellipsoid (based on J_ω) and the directions and lengths of the principal semi-axes of the three-dimensional linear force ellipsoid (based on J_v).

Exercise 5.27 Examine the software functions for this chapter in your favorite programming language. Verify that they work in the way that you expect. Can you make them more computationally efficient?